MIL-D-10692(Ord)
16 November 1950

MILITARY SPECIFICATION

Di- (2 ETHYL-HEXYL) SEBACATE

1. SCOPE

1.1 Scope.- This specification covers one type of di-(2 ethyl-hexyl) sebacate for use in polyisobutylene binder.

2. APPLICABLE SPECIFICATIONS

2.1 Specifications.- The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

FEDERAL SPECIFICATION

VV-L-791 - Lubricants, Liquid Fuels, and Related Products, Methods of Sampling and Testing.

U. S. ARMY SPECIFICATIONS

50-O-1 - General Specification for Ammunition Except Small Arms Ammunition.

100-2 - Standard Specification for Marking Shipments by Contractors.

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Color.- Colorless to amber.
3.2 Specific gravity, 20°C/4°C.- 0.913 ± 0.002.
3.3 Refractive index at 20°C.- 1.4495 ± 0.0010.
3.4 Flash point.- 400°F minimum.
3.5 Fire point.- 490°F minimum.
3.6 Acid number.- 0.1 maximum.
3.7 Viscosity at 100°F, poises.- 0.1153 ± 0.0010.
3.8 Saponification number.- 261 ± 2.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.- Unless otherwise specified in the contract or order, a lot shall consist of 50,000 pounds maximum, except that when shipment is made in tank cars, one car shall constitute a lot.

4.2 Sampling.-

4.2.1 Containers.- Ten percent, but in no case more than 10 or less than 3, of the containers shall be selected by the inspector so as to be representative of the lot. If there are fewer than 3 containers in the lot, all the containers shall be sampled. The material shall be mixed to a uniform consistency through-out and approximately 8 ounces of it shall be taken from each selected container. Each of these primary samples shall be placed in an airtight container and labeled so that the container from which it was taken can be identified. A composite sample of approximately 8 ounces shall be made from equal portions of the primary samples. The composite sample shall be thoroughly mixed and placed in an airtight container labeled to show the name of the material, manufacturer, plant, contract or order number, lot number and lot size. All acceptance tests shall be made on the composite sample. However, if it becomes apparent during sampling that the lot is not uniform, the inspector may require that any primary sample be tested for compliance with the requirements of the specification. All primary samples shall be held for possible future examination should the composite sample fail to meet the requirements.

4.2.2 Tank cars.- By means of a clean, weighed, small-neck, glass bottle approximately 1 quart of the material shall be removed. The unstopped bottle shall be rinsed with the material being sampled and lowered to the bottom of the tank by means of a cord, chain, or rod. The bottle shall be immediately withdrawn to the surface. In order to obtain a representative cross-section of the material, the speed of lowering and raising the bottle shall be uniform and so regulated that the bottle is just filled as it reaches the surface of the liquid. This composite sample shall be placed in a glass bottle having a glass stopper or cap containing a cork or cardboard liner covered with metal foil and labeled to show the name of the material, manufacturer, plant, contract or order number, lot number, and number of pounds in the lot. All acceptance tests shall be made on the composite sample representative of the lot.

4.3 Inspection.- Inspection shall be as specified in Specification 50-O-1.

4.4 Determination of color.-

4.4.1 Preparation of standard color solution.- Transfer 0.202 ± 0.001 gram of C.P. iodine and 1.000 ± 0.001 gram of C.P. potassium iodide to a 1000 ml volumetric flask. Add sufficient distilled water to dissolve the iodine and potassium iodide. After solution of the iodine and potassium iodide are complete, add sufficient distilled

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water to fill the volumetric flask to mark and mix thoroughly. The standard color solution shall not be used after standing approximately 24 hours.

4.4.2 Procedure.- Fill a clean 50 ml Nessler tube to the mark with the sample. Fill another clean 50 ml Nessler tube to the mark with the standard color solution described above. Hold the tube containing the sample and the tube containing the standard color solution side by side, and compare their colors by transmitted light. Consider the sample to be satisfactory if the color is not darker than the standard.

4.5 Determination of specific gravity.- Determine the specific gravity of the sample at 20°C/4°C to the third decimal place with any suitable pycnometer.

4.6 Determination of refractive index.- Determine the refractive index at 20°C with an Abbe Refractometer or any other equally accurate instrument.

4.7 Determination of flash point.- Flash point shall be determined in accordance with Method No. 110.3 of Specification VV-L-791.

4.8 Determination of fire point.- Fire point shall be determined in accordance with Method No. 110.3 of Specification VV-L-791.

4.9 Determination of acid number.- Transfer approximately 100 ml of a solution consisting of equal parts by volume of 95 percent alcohol and reagent grade benzene to a 300 ml Erlenmeyer Flask. Heat the flask and contents to incipient boiling on a steam bath. Add 2 or 3 drops of approximately 1% phenolphthalein indicator to the flask and titrate contents of the flask with approximately 0.01 N alkali hydroxide solution of know normality to a faint pink color. Add 50 ml of sample and again titrate the contents of the flask to a faint pink color. Calculate as follows, the number of mg of potassium hydroxide required to neutralize 1 gm of sample from the number of mls of standard alkali required to titrate the acidity of the sample.

$$\text{Acid number} = \frac{56.1 \times A \times N}{B \times D}$$

where:

A = ml. of alkali hydroxide required for sample.
N = normality of alkali hydroxide.
B = volume of sample taken.
D = specific gravity of sample.

4.10 Determination of viscosity.- The viscosity shall be determined with the aid of a calibrated Ubbelohde Viscosimeter in accordance with Method A of Method 30.5 of Specification VV-L-791.

4.11 Determination of saponification number.- Transfer an accurately weighed portion of approximately 2 gm of sample to a 300 ml ground glass Erlenmeyer or round bottom flask. Add 50 ml of approximately

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0.5N alcoholic potassium hydroxide solution. Connect the flask to an air condenser approximately 250 ml in length and reflux for 3 hours on a steam bath at such a rate that the major portion of the alcohol will condense in the lower half of the condenser as indicated by a sharp drop in temperature in the lower 125 ml of the condenser. Permit the flask and contents to cool to room temperature.

Remove the condenser and rinse with three 10 ml portions of neutralized 95% alcohol collecting the washings in the flask. Add 2 or 3 drops of 1% phenolphthalein indicator and titrate with approximately 0.5N hydrochloric acid to the first permanent, faint pink color. Make a blank determination on 50 ml of alcoholic KOH in accordance with the procedure given above. Calculate the saponification number as follows:

$$\text{Saponification Number:} = \frac{(A - B)\ N \times 56.1}{W}$$

where:

A = ml of hydrochloric acid required for sample.
B = ml of hydrochloric acid required for Blank.
N = normality of hydrochloric acid.
W = weight of sample.

4.12 Resubmission and retest.- If the composite sample, or any primary sample subjected to tests, fails to pass the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on sample taken from all or any of the containers in the lot at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot for acceptance provided complete replacement of the defective portion can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted, provided new samples, selected in accordance with 4.2, pass all the tests required by this specification.

5. PREPARATION FOR DELIVERY

5.1 Packing.- Unless otherwise specified, di-(2 ethyl-hexyl) sebacate shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carriers for safe transportation, at the lowest rate, to the point of delivery.

5.2 Labeling and marking.- In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with Specification 100-2.

NOTICE:-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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